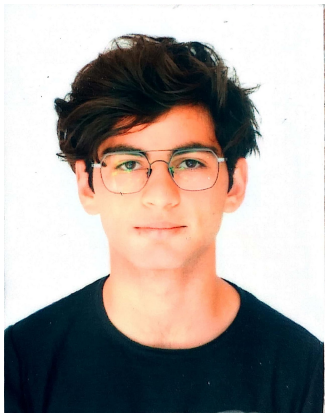
Emiliano Rizza

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Education

Jagiellonian University — Krakow (Poland)
currently Ph.D. student in Theoretical Physics — 06/2023–present
- Project: "Quantum information aspects of holography"
- Supervisors: Prof. Romuald Janik, Dr. Mario Flory

University of Florence — Florence (Italy)
Erasmus + — 10/2024–07/2025
- Project: "Rényi second laws for charged AdS black holes"
- Supervisor: Prof. Alice Bernamonti, Dr. Federico Galli

University of Catania — Catania (Italy)
M.S. in Theoretical Physics, 110/110 with honours — 09/2020–07/2022
- Thesis: "Lifshitz-type quantum field theories"
- Supervisor: Prof. Dario Zappalà

University of Catania — Catania (Italy)
B.S. in Physics and Astronomy, 110/110 with honours — 09/2017–12/2020
- Thesis: "Adiabatic Dynamics and Application to Quantum Information"
- Supervisor: Prof. Giuseppe Falci

Publications

A. Bhattacharya, M. Flory, M. P. Heller, E. Rizza, T. Schuhmann, "**The quadratic growth of Krylov spread complexity in the BTZ black hole**"
arXiv:2608.09922 — *Submitted: 10 August 2026*

E. Rizza, D. Zappalà, "**Lorentz symmetry violating Lifshitz-type field theories**",
Modern Physics Letters A, Vol. 37, No. 30
Journal / DOI | arXiv:2209.11060 — *Published: 30 December 2022*

Conferences and Schools – Presentations

9th Holography, String Theory and Observational Cosmology — *24–28 August 2026*
Kantary Hills Hotel, Chiang Mai, Thailand — *Invited Speaker*
Talk: "The quadratic growth of Krylov spread complexity in the BTZ black hole"

EUROSTRINGS 2026 – COST Action Theory-Challenges General Meeting — *4–7 May 2026*
National and Kapodistrian University of Athens, Athens, Greece — *Poster Presentation*

Poster: "Rényi second laws for charged AdS black holes"

89th Annual Meeting of the DPG Matter and Cosmos Section *15–20 March 2026*
Erlangen, Germany *Contributed Talk*
Talk: "Rényi second laws for charged AdS black holes"

XXVII Giambiagi Winter School – Holography, Black Holes and Information Theory *21–26 July 2025*
FCEyN, University of Buenos Aires, Buenos Aires, Argentina *Poster Presentation*
Poster: "Rényi second laws for charged AdS black holes"

First Kraków–Paderborn Quantum-Information Mini-Symposium *8–12 July 2024*
Jagiellonian University, Kraków, Poland *Contributed Talk / Organiser*
Talk: "Maximum Entropy Principle in Quantum Mechanics, Gravity and Complexity"

Cracow School of Theoretical Physics *15–23 June 2024*
From the UltraViolet to the InfraRed: A Panorama of Modern Gravitational Physics
Zakopane, Tatra Mountains, Poland *Contributed Talk*
Talk: "Lorentz symmetry violating Lifshitz-type field theories"

DPG Spring Meeting of the Condensed Matter Section *17–22 March 2024*
Technical University of Berlin, Berlin, Germany *Contributed Talk*
Talk: "Lorentz symmetry violating Lifshitz-type field theories"

Conferences and Schools – Participation

Iberian Strings'25 *15–17 January 2025*
University of the Basque Country, Bilbao, Spain

LACES 2024 – Lezioni Avanzate di Campi e Stringhe *25 November–13 December 2024*
Galileo Galilei Institute for Theoretical Physics, Florence, Italy

stringtheory.pl/2024 *7–8 June 2024*
Jagiellonian University, Kraków, Poland

9th Conference of the Polish Society on Relativity *18–22 September 2023*
Jagiellonian University, Kraków, Poland

29th WE-Heraeus Summer School "Saalburg" *4–15 September 2023*
Bayrischzell, Bavaria, Germany

Quantum Entanglement in High Energy Physics *10–13 May 2023*
Jagiellonian University, Kraków, Poland

Honours and Awards

- **Italian Olympics in Physics "OLIFIS"**
  Awarded with 3rd place 2017-2016
  Liceo Scientifico "Medi", Senigallia, (IT)

Working Experience

- High school teacher on fixed-term contract to teach Mathematics and Physics subjects 10/2022–12/2022
  Liceo Scientifico Pietro Farinato, Enna (IT)
- Teacher training course for the acquisition of 24 ECTS
  in the anthropological-psycho-pedagogical fields and in teaching methodologies 08/2022

eCampus University, Catania (IT)

Skills

- **Digital:** Python, C, LateX, Matlab, Mathematica
- **Communication:** Excellent communication skills confirmed through the experience as a tutor in the preparation course for the Italian Olympics in Physics and as a highschool teacher

Languages

- **English:** Proficiency
  - **IELTS:** score 7.0
- **Italian:** Native Speaker

Research Grants and Scholarships

Research Support Module *Awarded 20 March 2026*
Project: "Krylov spread complexity and the ASEP model"
Research mobility grant for a one-month visiting research stay at the University of Tokyo.

Research Support Module *Awarded 21 October 2025*
Project: "Krylov spread complexity in the DSSYK model"
Research equipment grant for computational studies of Krylov complexity and holography.

Extracurricular Activities

- "Numerical Methods in Financial Physics" Course 2026
  Jagiellonian University, Krakow (PL)
- "Introduction to Random Matrix and Quantum Integrable Models" Course 2025
  Università di Firenze, Firenze (IT)
- "Numerical calculations using Mathematica" Course 2024
  Jagiellonian University, Krakow (PL)
- "Introduction to conformed field theories" Course 2024
  Jagiellonian University, Krakow (PL)
- "Introduction to Quantum Gravity" Course 2024
  Jagiellonian University, Krakow (PL)
- "Relativistic hydrodynamics" Course 2024
  Jagiellonian University, Krakow (PL)